

Definite Integration

21. Let (a, b) be the point of intersection of the curve $x^2 = 2y$ and the straight line $y - 2x - 6 = 0$ in the second quadrant.

Then the integral $I = \int_a^b \frac{9x^2}{1+5^x} dx$ is equal to :

माना वक्र $x^2 = 2y$ और सरल रेखा $y - 2x - 6 = 0$ का दूसरे चतुर्थांश में प्रतिच्छेदन बिन्दु (a, b) है। तो समाकलन $I = \int_a^b \frac{9x^2}{1+5^x} dx$

बराबर है :

(A) 24

(B) 18

(C) 21

(D) 27

Ans. A

Sol. $x^2 = 2y$ and $y - 2x - 6 = 0$

$$\frac{x^2}{2} - 2x - 6 = 0$$

$$x^2 - 4x - 12 = 0$$

$$x^2 - 6x + 2x - 12 = 0$$

$$x(x - 6) + 2(x - 6) = 0$$

$$(x - 6)(x + 2) = 0$$

Point of intersection are (6, 18) and (-2, 2)

(-2, 2) is in second quadrant

$$a = -2, b = 2$$

$$I = \int_{-2}^2 \frac{9x^2}{1+5^x} dx \quad \dots(1)$$

$$I = \int_{-2}^2 \frac{9x^2}{1+5^{-x}} dx \quad \dots(2)$$

Adding (1) and (2)

$$2I = \int_{-2}^2 9x^2 dx$$

$$I = 9 \int_0^2 x^2 dx$$

$$I = 9 \left(\frac{x^3}{3} \right)_0^2 \Rightarrow I = 24$$

22. $4 \int_0^1 \left(\frac{1}{\sqrt{3+x^2} + \sqrt{1+x^2}} \right) dx - 3 \log_e (\sqrt{3})$ is equal to :

$4 \int_0^1 \left(\frac{1}{\sqrt{3+x^2} + \sqrt{1+x^2}} \right) dx - 3 \log_e (\sqrt{3})$ बराबर है :

- (A) $2 + \sqrt{2} - \log_e (1 + \sqrt{2})$ (B) $2 + \sqrt{2} + \log_e (1 + \sqrt{2})$
 (C) $2 - \sqrt{2} + \log_e (1 + \sqrt{2})$ (D) $2 - \sqrt{2} - \log_e (1 + \sqrt{2})$

Ans. D

Sol. $I = 4 \int_0^1 \frac{1}{\sqrt{3+x^2} + \sqrt{1+x^2}} dx$
 $= 2 \int_0^1 \sqrt{3+x^2} - \sqrt{1+x^2} dx$
 $= 2 \left[\int_0^1 \sqrt{3+x^2} dx - \int_0^1 \sqrt{1+x^2} dx \right]$
 $= 2 \left[\left(\frac{1}{2} x \sqrt{x^2+3} + \frac{3}{2} \ln |\sqrt{x^2+3} + x| \right) - \left(\frac{1}{2} x \sqrt{1+x^2} + \frac{1}{2} \ln |\sqrt{1+x^2} + x| \right) \right]_0^1$
 $= 2 \left[\left(1 + \frac{3}{2} \ln 3 - \frac{3}{2} \ln \sqrt{3} \right) - \left(\frac{\sqrt{2}}{2} + \frac{1}{2} \ln (\sqrt{2} + 1) \right) \right]$
 $= 2 \left(1 + \frac{3}{4} \ln 3 - \frac{1}{\sqrt{2}} - \frac{1}{2} \ln (\sqrt{2} + 1) \right)$
 $= 3 \ln \sqrt{3} + 2 - \sqrt{2} - \ln (\sqrt{2} + 1)$
 $I - 3 \ln \sqrt{3} = 2 - \sqrt{2} - \ln (\sqrt{2} + 1)$

23. Let the domain of the function $f(x) = \log_2 \log_4 \log_6 (3 + 4x - x^2)$ be (a, b) . If $\int_0^{b-a} [x^2] dx = p - \sqrt{q} - \sqrt{r}$, $p, q, r \in \mathbb{N}$, $\gcd(p, q, r) = 1$, where $[\cdot]$ is the greatest integer function, then $p + q + r$ is equal to :

माना फलन $f(x) = \log_2 \log_4 \log_6 (3 + 4x - x^2)$ का प्रांत (a, b) है। यदि $\int_0^{b-a} [x^2] dx = p - \sqrt{q} - \sqrt{r}$, $p, q, r \in \mathbb{N}$, $\gcd(p, q, r) = 1$ हैं, जहाँ $[\cdot]$ अधिकतम पूर्णांक फलन है, तो $p + q + r$ बराबर है :

(A) 11

(B) 10

(C) 8

(D) 9

Ans. B

Sol. $\log_4 \log_6 (3 + 4x - x^2) > 0$

$$\log_6 (3 + 4x - x^2) > 1$$

$$3 + 4x - x^2 > 6$$

$$x \in (1, 3)$$

$$a = 1, b = 3$$

$$I = \int_0^2 [x^2] dx = \int_0^1 [x^2] dx + \int_1^{\sqrt{2}} [x^2] dx + \int_{\sqrt{2}}^{\sqrt{3}} [x^2] dx + \int_{\sqrt{3}}^2 [x^2] dx$$

$$= 5 - \sqrt{2} - \sqrt{3}$$

$$p + q + r = 10$$

24. The integral $\int_0^{\pi} \frac{8x dx}{4 \cos^2 x + \sin^2 x}$ is equal to :

समाकलन $\int_0^{\pi} \frac{8x dx}{4 \cos^2 x + \sin^2 x}$ बराबर है :

(A) $\frac{3\pi^2}{2}$

(B) π^2

(C) $4\pi^2$

(D) $2\pi^2$

Ans. D

Sol. $\therefore I = \int_0^{\pi} \frac{8x}{4 \cos^2 x + \sin^2 x} dx \quad \dots(1)$

$$I = \int_0^{\pi} \frac{8(\pi - x)}{4 \cos^2 x + \sin^2 x} dx \quad \dots(2)$$

(A) + (2)

$$2I = \int_0^{\pi} \frac{8\pi}{4 \cos^2 x + \sin^2 x} dx$$

$$I = 4\pi \int_0^{\pi} \frac{1}{1 + 3 \cos^2 x} dx$$

$$I = 8\pi \int_0^{\frac{\pi}{2}} \frac{1}{1+3\cos^2 x} dx$$

$$I = 8\pi \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{\sec^2 x + 3} dx$$

$$I = 8\pi \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{4 + \tan^2 x} dx$$

Put

$$\tan x = t$$

$$\sec^2 x dx = dt$$

$$I = \int_0^{\infty} \frac{dt}{t^2 + 2^2}$$

$$I = 4\pi \left(\tan^{-1} \left(\frac{t}{2} \right) \right)_0^{\infty} \Rightarrow U = 4\pi \times \frac{\pi}{2} = 2\pi^2$$

25. The value of $\int_{-1}^1 \frac{(1 + \sqrt{|x| - x})e^x + (\sqrt{|x| - x})e^{-x}}{e^x + e^{-x}} dx$ is equal to :

$$\int_{-1}^1 \frac{(1 + \sqrt{|x| - x})e^x + (\sqrt{|x| - x})e^{-x}}{e^x + e^{-x}} dx \text{ का मान बराबर होगा ।}$$

$$(A) 2 + \frac{2\sqrt{2}}{3}$$

$$(B) 3 - \frac{2\sqrt{2}}{3}$$

$$(C) 1 - \frac{2\sqrt{2}}{3}$$

$$(D) 1 + \frac{2\sqrt{2}}{3}$$

Ans. D

Sol. $I = \int_{-1}^1 \frac{(1 + \sqrt{|x| - x})e^x + (\sqrt{|x| - x})e^{-x}}{e^x + e^{-x}} dx \quad \dots(1)$

Apply King's Property

$$I = \int_{-1}^1 \frac{(1 + \sqrt{|x| + x})e^{-x} + (\sqrt{|x| + x})e^x}{e^{-x} + e^x} dx \quad \dots(2)$$

Add (1) & (2)

$$2I = \int_{-1}^1 \frac{e^x \{1 + \sqrt{|x| - x} + \sqrt{|x| + x}\} + e^{-x} \{\sqrt{|x| - x} + 1 + \sqrt{|x| + x}\}}{e^x + e^{-x}} dx$$

$$= \int_{-1}^1 \frac{(1 + \sqrt{|x| - x} + \sqrt{|x| + x})(e^x + e^{-x})}{e^x + e^{-x}} dx$$

$$2I = \int_{-1}^1 (1 + \sqrt{|x| - x} + \sqrt{|x| + x}) dx$$

Even function

$$I = \int_0^1 (1 + \sqrt{|x| - x} + \sqrt{|x| + x}) dx$$

$$= \int_0^1 (1 + \sqrt{2x}) dx$$

$$I = 1 + \sqrt{2} \cdot \frac{2x^{\frac{3}{2}}}{3} \Big|_0^1$$

$$= 1 + \frac{2\sqrt{2}}{3}$$

26. Let $f(x) + 2f\left(\frac{1}{x}\right) = x^2 + 5$ and $2g(x) - 3g\left(\frac{1}{2}\right) = x, x > 0$. If $\alpha = \int_1^2 f(x)dx$, and $\beta = \int_1^2 g(x)dx$, then the value of $9\alpha + \beta$ is :

माना $f(x) + 2f\left(\frac{1}{x}\right) = x^2 + 5$ और $2g(x) - 3g\left(\frac{1}{2}\right) = x, x > 0$ हैं। यदि $\alpha = \int_1^2 f(x)dx$, और $\beta = \int_1^2 g(x)dx$ हैं, तो $9\alpha + \beta$ का मान है :

- (A) 0 (B) 10 (C) 11 (D) 1

Ans. C

Sol. $f(x) + 2f\left(\frac{1}{x}\right) = x^2 + 5$

$$2f(x) + f\left(\frac{1}{x}\right) = \frac{1}{x^2} + 5 \Big\} \times 2$$

$$-3f(x) = x^2 + 5 - \frac{2}{x^2} - 10$$

$$f(x) = \frac{-1}{3} \left(x^2 - \frac{2}{x^2} - 5 \right)$$

$$\alpha = \frac{-1}{3} \left(\frac{x^3}{3} + \frac{2}{x} - 5x \right)_1$$

$$= -\frac{1}{3} \left(\left(\frac{8}{3} + 1 - 10 \right) - \left(\frac{1}{3} + 2 - 5 \right) \right)$$

$$= \frac{-1}{3} \left(\frac{7}{3} - 6 \right) = \frac{-1}{3} \left(\frac{-11}{3} \right) = \frac{11}{9}$$

$$\Rightarrow 2g(x) - 3g\left(\frac{1}{2}\right) = x$$

$$\text{Put } x = \frac{1}{2} \Rightarrow g\left(\frac{1}{2}\right) = \frac{-1}{2}$$

$$2g(x) = x - \frac{3}{2}$$

$$g(x) = \frac{1}{2} \left(x - \frac{3}{2} \right)$$

$$\beta = \frac{1}{2} \left(\frac{x^2}{2} - \frac{3}{2}x \right)_1$$

$$= \frac{1}{2} \left((2 - 3) - \left(\frac{1}{2} - \frac{3}{2} \right) \right)$$

$$= \frac{1}{2} ((-1) - (-1)) = 0$$

27. The integral $\int_0^\pi \frac{(x+3)\sin x}{1+3\cos^2 x} dx$ is equal to :

समाकलन $\int_0^\pi \frac{(x+3)\sin x}{1+3\cos^2 x} dx$ बराबर है :

(A) $\frac{\pi}{\sqrt{3}}(\pi+2)$

(B) $\frac{\pi}{\sqrt{3}}(\pi+1)$

(C) $\frac{\pi}{3\sqrt{3}}(\pi+6)$

(D) $\frac{\pi}{2\sqrt{3}}(\pi+4)$

Ans. C

Sol. $I = \int_0^{\pi} \frac{(x+3)\sin x}{1+3\cos^2 x} dx$

$$I = \int_0^{\pi} \frac{(\pi-x+3)\sin x}{(1+3\cos^2 x)} dx$$

$$2I = \int_0^{\pi} \frac{(\pi+6)\sin x dx}{(1+3\cos^2 x)} = 2 \int_0^{\frac{\pi}{2}} \frac{(\pi+6)\sin x}{(1+3\cos^2 x)}$$

$$I = \int_0^{\frac{\pi}{2}} \frac{(\pi+6)\sin x dx}{(1+3\cos^2 x)} = \frac{\pi}{3\sqrt{3}}(\pi+6)$$

$$\sqrt{3} \cos x = t$$

$$\sqrt{3} \sin x = dt$$

28. Let $f(x)$ be a positive function and $I_1 = \int_{-\frac{1}{2}}^1 2xf(2x(1-2x))dx$ and $I_2 = \int_{-1}^2 f(x(1-x))dx$. Then the value of $\frac{I_2}{I_1}$ is equal to :

. माना $f(x)$ एक धनात्मक फलन है तथा $I_1 = \int_{-\frac{1}{2}}^1 2xf(2x(1-2x))dx$ और $I_2 = \int_{-1}^2 f(x(1-x))dx$ हैं, तो $\frac{I_2}{I_1}$ का मान बराबर है :

(A) 9

(B) 4

(C) 12

(D) 6

Ans. B

Sol. $I_1 = \int_{-1/2}^1 2xf(2x(1-2x))dx$

$$I_2 = \int_{-1}^2 t(f(t(1-t))) \frac{dt}{2} \quad \text{_____ (i)} \quad (\text{put } 2x=t)$$

Use prop. V

$$I_1 = \int_{-1}^2 (1-t)f(1-t)t \frac{dt}{2} \quad \text{_____ (ii)}$$

Equation (1) + (2)

$$2I_1 = \int_{-1}^2 f(t(1-t)) \frac{dt}{2}$$

$$4I_1 = \int_{-1}^2 f(t(1-t))dt = I_2$$

29. The integral $\int_{-1}^{\frac{3}{2}} (\pi^2 x \sin(\pi x)) dx$ is equal to :

समाकलन $\int_{-1}^{\frac{3}{2}} (\pi^2 x \sin(\pi x)) dx$ बराबर है :

- (A) $1 + 3\pi$ (B) $2 + 3\pi$ (C) $4 + \pi$ (D) $3 + 2\pi$

Ans. A

Sol. $I = \int_{-1}^{3/2} |\pi^2 x \sin \pi x| dx = \int_{-1}^{3/2} \pi |\pi x \sin \pi x| dx$

$$I = \int_{-\pi}^{3\pi/2} |t \sin t| dt \quad (\text{let } \pi x = t \Rightarrow \pi dx = dt)$$

$$I = 2 \int_0^{\pi} |t \sin t| dt + \int_{\pi}^{3\pi/2} |t \sin t| dt$$

$$I = 2 \int_0^{\pi} t \sin t dt - \int_{\pi}^{3\pi/2} t \sin t dt$$

$$I = 2 \left(\left(t(-\cos t) \right)_0^{\pi} - \int_0^{\pi} -\cos t dt \right) - \left(\left(t(-\cos t) \right)_{\pi}^{3\pi/2} - \int_{\pi}^{3\pi/2} -\cos t dt \right)$$

$$I = 2 \left(\pi \cdot 1 + (\sin t)_0^{\pi} \right) - \left(+\pi \cdot (-1) + (\sin t)_{\pi}^{3\pi/2} \right)$$

$$I = 2\pi + \pi + 1 = 1 + 3\pi$$

30. Let for $f(x) = 7 \tan^8 x + 7 \tan^6 x - 3 \tan^4 x - 3 \tan^2 x$, $I_1 = \int_0^{\pi/4} f(x) dx$ and $I_2 = \int_0^{\pi/4} x f(x) dx$. then $7I_1 + 12I_2$ is equal to :

माना कि $f(x) = 7 \tan^8 x + 7 \tan^6 x - 3 \tan^4 x - 3 \tan^2 x$ के लिए, $I_1 = \int_0^{\pi/4} f(x) dx$ और $I_2 = \int_0^{\pi/4} x f(x) dx$ है। तब $7I_1 + 12I_2$ बराबर है :

- (A) 2 (B) 2π (C) π (D) 1

Ans. D

Sol. $f(x) = (7 \tan^6 x - 3 \tan^2 x) \cdot \sec^2 x$

$$\therefore I_1 = \int_0^{\frac{\pi}{4}} f(x) dx = \int_0^1 (7t^6 - 3t^2) dt = (t^7 - t^3)_0^1 = 0$$

Now
$$I_2 = \int_0^{\frac{\pi}{4}} x f(x) dx = \int_0^1 \frac{(7t^6 - 3t^2)}{II} \frac{\tan^{-1} t}{I} dt$$

$$= (\tan^{-1} t (t^7 - t^3))_0^1 - \int_0^1 (t^7 - t^3) \cdot \frac{1}{1+t^2} dt$$

$$= \int_0^1 \frac{t^3(1-t^4)}{1+t^2} dt = \int_0^1 t^3(1-t^2) dt$$

$$= \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$$

Now, $7I_1 + 12I_2 = 1$

31. The value of $\int_{e^2}^{e^4} \frac{1}{x} \left(\frac{e^{((\log_e x)^2 + 1)^{-1}}}{e^{((\log_e x)^2 + 1)^{-1}} + e^{((6 - \log_e x)^2 + 1)^{-1}}} \right) dx$ is :

$$\int_{e^2}^{e^4} \frac{1}{x} \left(\frac{e^{((\log_e x)^2 + 1)^{-1}}}{e^{((\log_e x)^2 + 1)^{-1}} + e^{((6 - \log_e x)^2 + 1)^{-1}}} \right) dx \text{ का मान है :}$$

(A) 2

(B) $\log_e 2$

(C) e^2

(D) 1

Ans. D

Sol. put $\ell n x = t$

$$I = \int_2^4 \frac{e^{(t^2+1)^{-1}}}{e^{(t^2+1)^{-1}} + e^{((6-t)^2+1)^{-1}}} dt$$

King add

$$2I = \int_2^4 (1) dt$$

$$I = 1$$

32. If $I(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$, $m, n > 0$, then $I(9, 14) + I(10, 13)$ is :

यदि $I(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$, $m, n > 0$ है, तो $I(9, 14) + I(10, 13)$ है :

- (A) $I(9, 13)$ (B) $I(1, 13)$ (C) $I(19, 27)$ (D) $I(9, 1)$

Ans. A

Sol. $I(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$

Let $x = \sin^2 \theta$ $dx = 2 \sin \theta \cos \theta d\theta$

$$I(m, n) = 2 \int_0^{\frac{\pi}{2}} (\sin \theta)^{2m-1} (\cos \theta)^{2n-1} d\theta$$

$$I(9, 14) + I(10, 13) = 2 \int_0^{\frac{\pi}{2}} (\sin \theta)^{17} (\cos \theta)^{27} d\theta + 2 \int_0^{\frac{\pi}{2}} (\sin \theta)^{19} (\cos \theta)^{25} d\theta$$

$$= 2 \int_0^{\frac{\pi}{2}} (\sin \theta)^{17} (\cos \theta)^{25} d\theta$$

$$= I(9, 13)$$

33. If $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{96x^2 \cos^2 x}{(1+e^x)} dx = \pi(\alpha\pi^2 + \beta)$, $\alpha, \beta \in \mathbb{Z}$ then $(\alpha + \beta)^2$ equals :

यदि $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{96x^2 \cos^2 x}{(1+e^x)} dx = \pi(\alpha\pi^2 + \beta)$, $\alpha, \beta \in \mathbb{Z}$ है तो $(\alpha + \beta)^2$ बराबर है :

- (A) 196 (B) 64 (C) 144 (D) 100

Ans. D

Sol. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{96x^2 \cos^2 x}{(1+e^x)} dx$ (Apply king property)

$$\int_0^{\frac{\pi}{2}} 96x^2 \cos^2 x = 48 \int_0^{\frac{\pi}{2}} x^2 (1 + \cos 2x) dx$$

$$48 \left[\left(\frac{x^3}{3} \right)_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \frac{x^2}{I} \frac{\cos 2x}{II} dx \right]$$

$\Rightarrow$ On solving $\pi(2\pi^2 - 12)$

$\Rightarrow \alpha = 2, \beta = -12$

$\Rightarrow (\alpha + \beta)^2 = 100$

34. Let f be a real valued continuous function defined on the positive real axis such that

$g(x) = \int_0^x t f(t) dt$. If $g(x^3) = x^6 + x^7$, then value of $\sum_{r=1}^{15} f(r^3)$ is :

माना कि f धनात्मक वास्तविक अक्ष पर परिभाषित एक वास्तविक मानों का संतत फलन इस प्रकार है कि $g(x) = \int_0^x t f(t) dt$ है। यदि

$g(x^3) = x^6 + x^7$ है, तो $\sum_{r=1}^{15} f(r^3)$ का मान है :

(A) 340

(B) 320

(C) 270

(D) 310

Ans. D

Sol. $g(x) = \int_0^x t f(t) dt$

$$g'(x) = x \cdot f(x)$$

$$g(x^3) = x^6 + x^7$$

$$g'(x^3) \cdot 3x^2 = 6x^5 + 7x^6$$

$$x^3 \cdot f(x^3) \cdot 3x^2 = 6x^5 + 7x^6$$

$$3 \cdot f(x^3) = 6 + 7x$$

$$f(x^3) = \left(\frac{6 + 7x}{3} \right)$$

$$\sum_{r=1}^{15} f(r^3) = \sum_{r=1}^{15} \left(\frac{6 + 7r}{3} \right)$$

$$\Rightarrow 2 \left(\sum_{r=1}^{15} 1 \right) + \left(\frac{7}{3} \right) \left(\sum_{r=1}^{15} r \right)$$

$$\Rightarrow 2(15) + \left(\frac{7}{3} \right) \left(\frac{(15)(16)}{2} \right)$$

$$\Rightarrow 30 + 40(7)$$

$$\Rightarrow 30 + 280 = 310$$

35. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that $f(2) = 1$. If $F(x) = x f(x)$ for all $x \in \mathbb{R}$, $\int_0^2 x F'(x) dx = 6$ and $\int_0^2 x^2 F''(x) dx = 40$, then $F'(2) + \int_0^2 F(x) dx$ is equal to :

माना कि $f: \mathbb{R} \rightarrow \mathbb{R}$, एक दो बार अवकलनीय फलन इस प्रकार है कि $f(2) = 1$ है। यदि सभी $x \in \mathbb{R}$ के लिए, $F(x) = x f(x)$, $\int_0^2 x F'(x) dx = 6$ और $\int_0^2 x^2 F''(x) dx = 40$ है, तब $F'(2) + \int_0^2 F(x) dx$ बराबर है :

- (A) 11 (B) 15 (C) 13 (D) 9

Ans. A

Sol. $f(2) = 1$

$$F(x) = x f(x) \text{ (Given)}$$

$$\int_0^2 \underbrace{x F'(x)}_I dx = 6$$

Using by part

$$\left[\left(F'(x) \cdot \frac{x^2}{2} \right) - \int F''(x) \cdot \frac{x^2}{2} \right]_0^2 = 6$$

$$\Rightarrow \left[F'(x) \cdot \frac{x^2}{2} \right]_0^2 - \frac{\int_0^2 x^2 \cdot F''(x)}{2} = 6$$

$$\Rightarrow F'(2) \cdot (2) - \left(\frac{40}{2} \right) = 6$$

$$\Rightarrow F'(2) = 13$$

$$\Rightarrow F'(2) + \int_0^2 F(x) dx$$

$$\Rightarrow F'(2) + \left[F(x) \cdot x \right]_0^2 - \int_0^2 F'(x) \cdot x dx$$

$$\Rightarrow F'(2) + 2 \cdot F(2) - 6$$

$$\Rightarrow 13 + 2 \cdot (2) \cdot f(2) - 6$$

$$\Rightarrow 13 + (2) (2) (1) - 6$$

$$\Rightarrow 11$$

36. Let $f(x) = \int_0^x (t^2 - 9t + 20) dt, 1 \leq x \leq 5$. If the range of f is $[\alpha, \beta]$, then $4(\alpha + \beta)$ equals :

माना कि $f(x) = \int_0^x (t^2 - 9t + 20) dt, 1 \leq x \leq 5$ है। यदि f का परिसर $[\alpha, \beta]$, है, तो $4(\alpha + \beta)$ बराबर है :

(A) 157

(B) 253

(C) 154

(D) 125

Ans. A

Sol. $f'(x) = x^3 - 9x^2 + 20x = x(x-4)(x-5)$

$$\begin{array}{c} - \quad + \quad - \quad + \\ | \quad | \quad | \quad | \\ 0 \quad 4 \quad 5 \end{array}$$

$$\therefore f(x) = \frac{x^4}{4} - \frac{9x^3}{3} + \frac{20x^2}{2}$$

$$f(1) = \frac{1}{4} - 3 + 10 = \frac{29}{4} = \alpha$$

$$f(4) = \frac{256}{4} - 3(64)$$

$$f(4) = \frac{256}{4} - 3(64) + 10(16) = 32 = \beta$$

$$4(\alpha + \beta) = 4\left(\frac{29}{4} + 32\right) = 157$$

37. Let $f(x) = \begin{cases} -2, & -2 \leq x \leq 0 \\ x-2, & 0 < x \leq 2 \end{cases}$ and $h(x) = f(|x|) + |f(x)|$. Then $\int_{-2}^2 h(x) dx$ is equal to :

माना $f(x) = \begin{cases} -2, & -2 \leq x \leq 0 \\ x-2, & 0 < x \leq 2 \end{cases}$ तथा $h(x) = f(|x|) + |f(x)|$ हैं। तो $\int_{-2}^2 h(x) dx$ बराबर है :

(A) 2

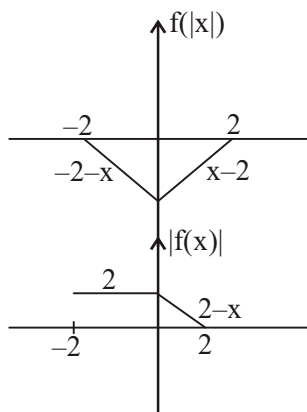
(B) 6

(C) 4

(D) 1

Ans. A

Sol.



$$\therefore h(x) = \begin{cases} -x & x \in [-2, 0] \\ 0 & x \in [0, 2] \end{cases}$$

$$\int_{-2}^2 h(x) dx = \int_{-2}^0 (-x) dx$$

$$= \left(-\frac{x^2}{2} \right)_{-2}^0$$

$$= -(-2) = 2$$

38. If the value of the integral $\int_{-1}^1 \frac{\cos \alpha x}{1+3^x} dx$ is $\frac{2}{\pi}$. Then, a value of α is :

यदि समाकलन $\int_{-1}^1 \frac{\cos \alpha x}{1+3^x} dx$ का मान $\frac{2}{\pi}$ है। तो α का एक मान है :

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$

(D) $\frac{\pi}{3}$

Ans. C

Sol. Let $I = \int_{-1}^{+1} \frac{\cos \alpha x}{1+3^x} dx$ _____ (1)

$$I = \int_{-1}^{+1} \frac{\cos \alpha x}{1+3^{-x}} dx$$

(using $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$) _____ (2)

Add (1) and (2)

$$2I = \int_{-1}^{+1} \cos(\alpha x) dx = 2 \int_0^1 \cos(\alpha x) dx$$

$$I = \frac{\sin \alpha}{\alpha} = \frac{2}{\pi} \text{ (given)}$$

$$\therefore \alpha = \frac{\pi}{2}$$

39. Let $f(x) = \int_0^x (t + \sin(1 - e^t)) dt$, $x \in \mathbb{R}$. Then, $\lim_{x \rightarrow 0} \frac{f(x)}{x^3}$ is equal to :

माना $f(x) = \int_0^x (t + \sin(1 - e^t)) dt$, $x \in \mathbb{R}$ है। तो $\lim_{x \rightarrow 0} \frac{f(x)}{x^3}$ बराबर है :

- (A) $\frac{1}{6}$ (B) $-\frac{1}{6}$ (C) $-\frac{2}{3}$ (D) $\frac{2}{3}$

Ans. B

Sol. $\lim_{x \rightarrow 0} \frac{f(x)}{x^3}$

Using L Hospital Rule.

$$\lim_{x \rightarrow 0} \frac{f'(x)}{3x^2} = \lim_{x \rightarrow 0} \frac{x + \sin(1 - e^x)}{3x^2} \text{ (Again L hospital)}$$

Using L Hospital Rule

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{-[\sin(1 - e^x)(-e^x) \cdot e^x + \cos(1 - e^x) \cdot e^x]}{6} \\ = -\frac{1}{6} \end{aligned}$$

40. The integral $\int_0^{\frac{\pi}{4}} \frac{136 \sin x}{3 \sin x + 5 \cos x} dx$ is equal to :

समाकलन $\int_0^{\frac{\pi}{4}} \frac{136 \sin x}{3 \sin x + 5 \cos x} dx$ बराबर है :

- (A) $3\pi - 30 \log_e 2 + 20 \log_e 5$ (B) $3\pi - 50 \log_e 2 + 20 \log_e 5$

(C) $3\pi - 10 \log_e (2\sqrt{2}) + 10 \log_e 5$

(D) $3\pi - 25 \log_e 2 + 10 \log_e 5$

Ans. B

Sol. $(136) \int_0^{\pi/4} \frac{\sin x}{3 \sin x + 5 \cos x} dx$

$$136 \sin x = \lambda (3 \sin x + 5 \cos x) + \mu (3 \cos x - 5 \sin x)$$

Compare coffi of $\sin x$, $\cos x$

$$136 = 3\lambda - 5\mu \quad \dots(1)$$

$$0 = 5\lambda + 3\mu \quad \dots(2)$$

$$408 = 34\lambda \Rightarrow \lambda = 12$$

$$\mu = -20$$

$$(136) \int_0^{\pi/4} \frac{\sin x}{3 \sin x + 5 \cos x} dx$$

$$= 12 \left(x \right)_0^{\pi/4} - 20 \left[\ln(3 \sin x + 5 \cos x) \right]_0^{\pi/4}$$

$$= 12 \frac{\pi}{4} - 20 \left[\ln 4\sqrt{2} - \ln 5 \right]$$

$$3\pi - 20 \ln(4\sqrt{2}) + 20 \ln 5$$

$$= 3\pi - 50 \ln 2 + 20 \ln 5$$